Inorganic Cumulative Exam
11-18-16

http://bit.ly/2fbpCZxSolid State :

Cadmium Selenide Nanoparticles have been explored extensively for over 20 years. These particles have been found to be fluorescent. The wavelength of fluorescence of cadmium selenide nanoparticles changes drastically depending on the preparation of the nanoparticles.

1) What is the stoichiometry of cadmium selenide nanoparticles?
2) What one property primarily determines the fluorescence wavelength of cadmium selenide nanoparticles? Explain your answer.

Superconductivity was first discovered in 1911. In the 1950's Bardeen, Cooper and Schrieffer described a theory explaining superconductivity.

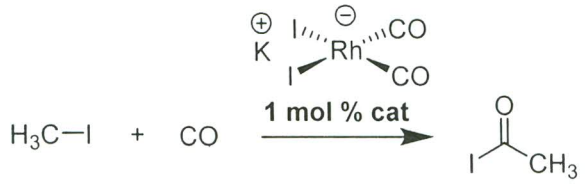
3) What experimental observations would lead you to conclude that your material is a superconductor?
4) Describe BCS theory.

Bulk semiconductors

5) Describe how p-doped silicon is produced.

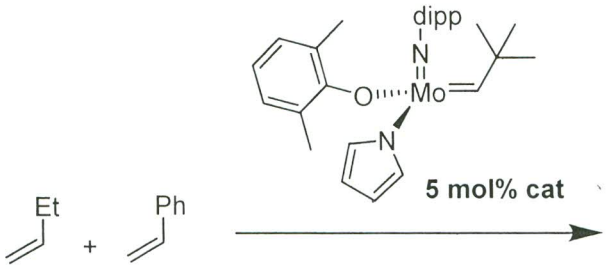
Homogeneous Chemistry and Theory:

The balanced reaction for part of the Monsanto Process is given below.



6) Provide a complete mechanism (including arrow pushing) for this portion of the Monsanto process.
7) Correctly classify each reaction in the mechanism you provided for this portion of the Monsanto process.

Provided below are the conditions for a reaction catalyzed by the chiral metal complex shown.



8) Provide ALL products (including isomers) of the above reaction. Do not omit any products that would be produced in small amounts (less than 1%).
9) Provide a mechanism for the production of one of the primary reaction products.

Electron transfer theory

10) Describe the difference between an inner sphere and outer sphere electron transfer reaction. Give an example of each type of reaction.